

How to Guide
SAP Business One 9.0 and Later
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CUSTOMER

Working with SAP Business One Studio Suite

All Countries

Typographic Conventions

Type Style	Description
<i>Example</i>	Words or characters quoted from the screen. These include field names, screen titles, pushbuttons labels, menu names, menu paths, and menu options. Textual cross-references to other documents.
Example	Emphasized words or expressions.
EXAMPLE	Technical names of system objects. These include report names, program names, transaction codes, table names, and key concepts of a programming language when they are surrounded by body text, for example, SELECT and INCLUDE.
Example	Output on the screen. This includes file and directory names and their paths, messages, names of variables and parameters, source text, and names of installation, upgrade and database tools.
Example	Exact user entry. These are words or characters that you enter in the system exactly as they appear in the documentation.
<Example>	Variable user entry. Angle brackets indicate that you replace these words and characters with appropriate entries to make entries in the system.
EXAMPLE	Keys on the keyboard, for example, F2 or ENTER .

Document History

Version	Date	Change
1.0	2012-11-05	The first version.
2.0	2013-01-22	Updated with 9.0 SP01.
3.0	2014-04-17	Updated with 9.1.
3.1	2015-10-16	Updated with 9.1 PL08.
4.0	2015-11-02	Updated with 9.2.
5.0	2017-08-25	Supports Visual Studio 2015 and TFS (Team Foundation Server) 2015.
6.0	2020-08-20	Supports Visual Studio 2017 and 2019.
6.1	2023-07-14	Update login methods when adding UDO forms

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1 Introduction

SAP Business One Studio Suite is a state-of-the-art development platform for SAP Business One extensions that dramatically improves partner development efficiency. It provides an effective integrated development environment (IDE) for partners to develop extension on top of SAP Business One. SAP Business One Studio Suite consists of two editions: SAP Business One Studio and SAP Business One Studio for Microsoft Visual Studio.

SAP Business One Studio is an open platform based on the .NET framework for a variety of designers. Currently, this platform supports two types of designer: Add-On Designer and Workflow Designer. In this document, SAP Business One Studio stands for SAP Business One Studio as add-on designer. You can use SAP Business One Studio to open SAP Business One system forms in edit mode, modify the existing controls and logic, or add new controls in forms. You can also design your own forms. SAP Business One Studio not only contains all the functionalities of ScreenPainter, but also enhances the functionalities and optimizes the usability of ScreenPainter.

SAP Business One Studio for Microsoft Visual Studio is a deeply customized programming environment for SAP Business One. Besides the functionalities of SAP Business One Studio, SAP Business One Studio for Microsoft Visual Studio leverages the powerful capability of Microsoft Visual Studio in development functionalities such as code generation, project and item template wizard, code intelligence, friendly GUI, and so on. What's more, the add-on project generated by SAP Business One Studio for Microsoft Visual Studio is based on new object level event framework of SAP Business One SDK. Together with system form editing, SAP Business One Studio for Microsoft Visual Studio makes it much easier to write a typical SAP Business One add-on, and the code generated is much simpler and maintainable.

SAP Business One Studio and SAP Business One Studio for Microsoft Visual Studio share with the same file format, .b1s, which is a package for all layout files and other relevant files. For using the two editions, a typical scenario is that a UI designer works on SAP Business One Studio for GUI drawing and polishing, and then a consultant works on it for customization. Afterwards, a developer can open the .b1s generated in SAP Business One Studio by SAP Business One Studio for Microsoft Visual Studio to program on it, and finish the complicated logic.

1.1 Installing SAP Business One Studio

Prerequisites

- The installation computer complies with all hardware and software requirements. For information on hardware and software requirements, refer to [SAP Help Portal](#).
- You have installed Microsoft .NET Framework 4.0.
- If you want to install SAP Business One Studio for Microsoft Visual Studio, you must install Microsoft Visual Studio 2010, 2012, 2013, 2015, TFS (Team Foundation Server) 2015, 2017 or 2019.

Note

If you are just designing your own UI layout, you do not need to install the SAP Business One client.

Procedure

To install SAP Business One Studio, perform the following steps:

1. Insert the SAP Business One Installation DVD and wait for the installation program to appear.
2. In the installation window, select the *Client* radio button.
3. In the *Client Components* window, select *Optional Components* and *SAP Business One Studio*.
4. Choose the *Install* button.
5. In the *Welcome* window, choose the *Next* button.
6. In the *Customer Information* window, specify the user name and company name.
7. In the *Setup Type* window, specify a setup type: *Complete* or *Custom*.

If you choose *Custom* type:

In the *Choose Destination Location* window: to accept the default location, choose the *Next* button; to select a different location, choose the *Browse* button.

In the *Select Features* window, select *SAP Business One Studio* and / or *SAP Business One Studio for Microsoft Visual Studio*, and then choose the *Next* button.

8. In the *Ready to Install the Program* window, choose the *Install* button.
9. In the *Setup Status* window, the installation wizard performs the required system actions.
10. In the *Complete* window, choose the *Finish* button.

Note

Alternatively, you can install the SAP Business One Studio application from the DVD folders:
Packages / B1Studio / setup.exe.

1.2 Running SAP Business One Studio

To run SAP Business One Studio, choose *All Programs* → *SAP Business One* → *SAP Business One Studio*, and choose *SAP Business One Studio* or *SAP Business One Studio for Microsoft Visual Studio*.

Alternatively, if you have an SAP Business One client installed, in SAP Business One, from the *Tools* menu, choose *SAP Business One Studio*, and choose *SAP Business One Studio* or *SAP Business One Studio for Microsoft Visual Studio*.

1.3 Terms and Definitions

The following terms are used in this document.

Term	Definition
B1s	SAP Business One solution file

Term	Definition
Screen Painter	An SAP Business One add-on. You can use Screen Painter to create user forms. The format of files generated by Screen Painter is .srf.
UI API	SAP Business One SDK User Interface Application Programming Interface
UDO Form	A form that is generated by the <i>User-Defined Object Registration</i> wizard.

1.4 Related Documentation

The documents listed in the table are referred to in this document.

Document	Location
User Interface Standards and Guidelines	http://service.sap.com/smb/sbocustomer/documentation and choose Release Family 9.0 → SDK and Custom Development
SDK online help file SDK_EN.CHM	http://service.sap.com/smb/sbocustomer/documentation and choose Release Family 9.0 → SDK and Custom Development Available also on the SAP Business One product DVD and in the download package from SAP Service Marketplace
SAP Business One Administrator's Guide	http://service.sap.com/smb/sbocustomer/documentation and choose Release Family 9.0 → System Administration Available also on the SAP Business One product DVD and in the download package from SAP Service Marketplace

2 SAP Business One Studio Main Window

You can access all SAP Business One Studio functions from the SAP Business One Studio main window, which is divided into the following areas:

- **Menu** button – located at the top of the SAP Business One Studio main window and provides some menu commands. For more information, see [Menu Button](#).
- **Toolbar** – located below the **Menu** button, and provides the icons for the commonly used functions. For more information, see [Toolbar](#).
- **Design area** – located in the middle of the SAP Business One Studio main window, and you can use this area to design your own forms. For more information, see [Designing a Form](#).
- **Tool windows:**
 - o **Toolbox** window – contains the controls you can use to design your form. For more information, see [Toolbox Window](#).
 - o **UI Outline** window – allows you to view items in a selected pane level and pane level 0. For example, when you select 2, the application displays items in pane level 2 and pane level 0.
 - o **Solution Explorer** window – allows you to view the forms in your solution. It contains three levels of items: solution level, project level and form level.
 - o **Properties** window – allows you to set and examine the properties of forms and form items.

Note

You can increase the viewing and editing space for the design area, depending on how you arrange the tool windows ([Toolbox](#) window, [UI Outline](#) window, [Solution Explorer](#) window and [Properties](#) window).

You can drag the border of the windows to resize the window, click  ([Auto Hide](#)) to hide the window, and also dock the windows to a new location (see [Docking Tool Windows](#)).

2.1 Menu Button

The SAP Business One Studio **Menu** button is located at the top of the SAP Business One Studio main window and contains the following menu commands:

Menu Command	Description / Activity
New	Creates a new add-on project or a workflow project.
Open	Opens an existing solution/project or a file.
Save Current File (Ctrl + S)	Saves the current file. B1s is refreshed to latest.
Export Current File As (Ctrl + Shift + E)	Exports the current file into the specified path. The file type is determined according to the project type. For example, you can export an add-on project file as a .srf file.

Menu Command	Description / Activity
<i>Save All</i> (C tr L + S h i ft + S)	Saves all changes to the project.
<i>Recent Files</i>	Shows the recently opened files.
<i>Recent Solutions</i>	Shows the recently opened solutions.
<i>View Start Page</i>	Opens or hides <i>Start Page</i> .
<i>Exit</i> (A lt + F 4)	Closes the SAP Business One Studio application.

2.2 Toolbar

The toolbar, located below the *Menu* button, is a collection of icons that provide easy access to commonly used functions. Active functions are shown in color, while inactive ones are grayed out.

Toolbar Icon	Description / Activity
 (<i>Undo</i> (C tr L + Z))	Reverses the most recent successful action.
 (<i>Redo</i> (C tr L + Y))	Redoes the last action that was undone.
 (<i>Copy</i> (C tr L + C))	Copies the selected control or text onto the clipboard.
 (<i>Cut</i> (C tr L + X))	Removes the selected control or text and places it on the clipboard.
 (<i>Paste</i> (C tr L + V))	Inserts the contents of the clipboard at the current location.
 (<i>Align Left</i>)	Aligns the left edges of the selected controls with the left edge of the last selected control.
 (<i>Align Center</i>)	Vertically aligns the selected controls with the center of the last selected control.
 (<i>Align Right</i>)	Aligns the right edges of the selected controls with the right edge of the last selected control.
 (<i>Align Top</i>)	Aligns the top edges of the selected controls with the top edge of the last selected control.
 (<i>Align Bottom</i>)	Aligns the bottom edges of the selected controls with the bottom edge of the last selected control.
 (<i>Align Middle</i>)	Horizontally aligns the selected controls with the center of the last selected control.
 (<i>Scale Size</i>)	Aligns the size of the selected controls with the last selected control.
 (<i>Scale Height</i>)	Aligns the height of the selected controls with the last selected control.
 (<i>Scale Width</i>)	Aligns the width of the selected controls with the last selected control.

Toolbar Icon	Description / Activity
 (Preview in SAP Business One)	Enables you to preview the form in the SAP Business One client.
 (Settings)	Enables you to select language and theme style of the user interface. There are 2 options of theme style: "Nova" has the same look as that for SAP Business One release 8.82; "NS" has the same look as that for SAP Business One 9.0 and later. To apply the new settings, choose Update , and restart SAP Business One Studio.

2.3 Toolbox Window

The [Toolbox](#) window is a collection of controls that you can use to design your form. It contains [UI Controls](#) and [Data Sources](#). To add a control, double-click it or drag and drop it into the design area:

- The newly added UI controls appear in the form; you can modify the position, size, and other specific properties of the controls in the [Properties](#) window.
- The newly added data sources appear at the bottom of the design area; you can modify the properties in the [Properties](#) window.

UI Controls

UI Control	Description / Activity
StaticText	Represents a label item. Specific Property: Caption - the text to display in the label.
EditText	Represents an edit text box. Specific Properties: • DataSource - the data source bound to the edit text box. • ChooseFromListUID - the unique ID of the ChooseFromList object to attach with this edit text box. • ChooseFromListAlias - the database field by which to filter a ChooseFromList attached to the edit box. When you enter a value in the text box and then press the tab key, the ChooseFromList dialog is displayed with only those rows where the value of the field specified in this property starts with the entered text. After you select a value, the key of the selected row is entered into the edit text box. • IsPassword - specifies whether the input data is of the same look and feel as a password field. • SuppressZeros - indicates whether to hide leading zeros as well as trailing zeros after a decimal point.

UI Control	Description / Activity
	• TabOrder - the order in which this edit text gets focus when you press the <code>TAB</code> key.
ExtendedEditText	Represents a text box that can display multiple lines.
Button	Represents a button. Specific Properties: • ButtonType - indicates whether the button acts as a text button or an image button. • Caption - the text to display on the button. • ChooseFromListUID - the unique ID of the ChooseFromList object to attach with this button. • Image - the path to an image displayed on the button.
CheckBox	Represents a check box.
OptionButton	Represents an option button (radio button).
ComboBox	Represents a combo box. Specific Properties: • DataSource - the data source bound to the combo box. • TabOrder - the order in which the combo box items get focus when you press the <code>TAB</code> key. • ValidValues - the valid values for this combo box item. Add or remove valid values from the popup window.
TabControl	Represents a tab control, which is a container of the items to be shown on the tab page. In UI API, the tab control is named as folder. Specific properties for tab pages: • AutoPaneSelection - indicates whether the tab control allows automatic pane selection. If you specify <code>True</code>, when you click on a tab, the application automatically sets the form pane level to the pane level bound to the tab control. • Caption - the text to display on the tab page. • Pane - the pane level of the tab. For more information, see Working with Tab Controls.
Grid	Represents a grid, which is a visual representation of a data table. Specific Properties: • CollapseLevel - the number of columns grouped together, starting from the left-most column. For example, if you specify 2 for this property, all rows are grouped by the values in the first column, and all rows within each group are grouped by the values in the second column. Default value is 0, indicating not collapsed. • Columns - the grid's columns. • DataTableID - the data table bound to the grid.

UI Control	Description / Activity
	SelectionMode - indicates how rows can be selected. For more information, see Working with Grids.
Matrix	Represents a matrix, which displays data in a table. Specific Properties: Columns - the collection of columns in the matrix. You can add or remove columns in the CollectionEditorForm window. MatrixLayout - the layout of the matrix. SelectionMode - indicates how rows can be selected. TabOrder - the order in which the items get focus when you press the <code>TAB</code> key. For more information, see Working with Matrixes.
PictureBox	Represents a picture box. Specific Property: Picture - the picture to display in the picture box.
LinkedTextBox	Represents an arrow link (➡). A linked button enables quick navigation between relevant objects. For example, a linked button next to the Customer field in the A/R Invoice form opens the Business Partner Master Data form for the relevant customer. Specific Properties: LinkTo - the unique ID of the control to which the linked button links. LinkedObject - the target object type of the linked button.
ButtonCombo	Represents an item that provides the functionality of a combo box and a button in a single control.

Data Sources

Data Source	Description / Activity
DB Data Source	Represents a SAP Business One database table to be attached to a form. Specific Property: TableName - an SAP Business One database table, for example, OCRD.
User Data Source	Represents a container for item data that is not stored in the database. Specific Properties: DataType - the data type of the user data source. Size - the size of the user data source. UniqueID - the unique ID of the user data source.
Data Tables	Represents data in table form. Specific Properties: Type - the type of the data table, including Manual and Query types.

Data Source	Description / Activity
	• Columns - the columns of the Manual type data table. • Query - the specified condition (query) of the Query type data table. • UniqueID - the unique ID of the data table. For more information, see Working with Data Tables.
Choose From List	Represents a ChooseFromList form. Specific Properties: • MultiSelection - indicates whether you can select multiple items in the ChooseFromList form. • ObjectType - the type of the object from which to generate the valid values. • UniqueID - the unique ID of the ChooseFromList. For more information, see Working with Choose From Lists.

2.4 Docking Tool Windows

To dock the tool windows ([Toolbox](#) window, [UI Outline](#) window, [Solution Explorer](#) window, and [Properties](#) window) to a designated area, perform the following steps:

1. Select the window you want to dock.
2. Drag the window from its current location towards the middle of the SAP Business One main window.
A guide diamond appears.
3. When the window you are dragging reaches the position where you want to dock it, move the pointer over the corresponding portion of the guide diamond.
4. To dock the window in the position indicated, release the mouse button.

3 Working with Add-on Projects in SAP Business One Studio

This section introduces how to work with your add-on projects in SAP Business One Studio.

3.1 Creating an Add-on Project

You can create your new add-on forms with the same look and feel as the SAP Business One forms.

Procedure

1. In the SAP Business One Studio main window, choose the [Menu](#) button and choose [New](#) → [Project](#) → [Add-on](#).
Alternatively, on the [Start Page](#) of SAP Business One Studio, choose [New Add-on](#).
2. In the [New Project](#) window, specify the name and the location of your new project.
3. Specify the package name.

Note

By default, the package name is the same as the project name.

The extension of a package (SAP Business One add-on solution) file is b1s.

4. If you need to create a project that does not contain any forms, select the [Create Empty Project](#) checkbox.
5. Choose [OK](#).

A default layout named NewForm 1 appears in the design area of the SAP Business One Studio main window.

You can design your own form now. For more information, see [Designing a Form](#).

3.2 Opening an Add-on Project or an External File

You can open an existing add-on project or a .srf file which was created by Screen Painter.

Procedure

1. In the SAP Business One Studio main window, choose the [Menu](#) button and choose [Open](#) → [Solution/Project](#) or [Open](#) → [File](#).
Alternatively, on the [Start Page](#) of SAP Business One Studio, choose [Open Project](#) or [Open File](#).
2. Specify the existing project or file you want to open.
3. Choose [Open](#).

The solution hierarchy appears in the [Solution Explorer](#) window.

3.3 Designing a Form

In the design area of the SAP Business One Studio main window, you can design your own form. This procedure introduces the basic operation of designing a form.

Procedure

1. In the *Toolbox* window, double-click or drag and drop a form item from the *UI Controls* area or the *Data Source* area.
For example, double-click a button. A new button appears in the top left corner of the form.
2. Use drag and drop to position or resize the item on the form.
Note
When you drag the item, the application shows a red line which enables you to align the item with the existing ones.
3. Select the form or an item to set the properties.
You can select the form or a form item either from the design area or from the *UI Outline* window. The *UI Outline* window groups form items in pane levels.
4. In the *Properties* window, edit the properties.
5. To preview the form in the SAP Business One client, click  (*Preview@B1Client*).
6. Save the form.

3.3.1 Working with Tab Controls

A tab control is a container of the items to be shown on the tab page. In UI API, the tab control is named as folder.

Procedure

1. Add a tab control.
To add a tab control, in the *Toolbox* window, *UI Controls* area, double-click the *TabControl* icon.
The tab control is added to the form with one tab page.
If the tab control is active, a cross-shaped symbol  appears at the top left of the tab control.
2. To add or remove a tab page, right-click the cross-shaped symbol  and choose *Add Tab Page* or *Remove Tab Page*.
3. Specify properties for each tab page.
To specify the pane level of the tab page, in the *Properties* window, set the *AutoPaneSelection* field to True, and in the *Pane* field, enter the pane level.
4. Add UI controls to the tab page area, and specify the *FromPane*, *ToPane* properties.
Once the *FromPane*, *ToPane* properties are set, when you switch tab pages, the UI controls with the same pane level are displayed on the tab page.

3.3.2 Working with Grids

The grid control is a visual representation of a data table.

Procedure

1. Add a grid control.
To add a grid control, in the Toolbox window, UI Controls area, double-click the Grid icon.
The grid control is added to the form without columns.
2. Add a data table.
To add a data table, in the Toolbox window, Data Source area, double-click the Data Tables icon.
For more information, see Working with Data Tables.
3. Bind the data table to the grid as follows:
 1. Select the grid control you want to bind.
 2. In the *Properties* window, under the *Specific* area, in the *DataTableID* field, select a data table to which you want to bind.

Note

- o If the data table type is Query, the columns are not displayed in edit mode. To preview the columns, click  (*Preview@B1Client*).
 - o If the data table type is Manual, the columns are displayed in edit mode.
3. After binding a grid control to a data table of Manual type, you can set the *Columns* property. Perform the following steps:
 1. In the *Columns* property field, click the  button.
The *CollectionEditorForm* window appears.
 2. Select the column you want to edit from the list in the left area of the window.
 3. In the *Properties* area, specify the column type.

Note

To set a column with a link button, select *it_EDIT* as the column type, and in the *LinkObjectType* field, specify the type of the object you want to connect to.

3.3.3 Working with Data Tables

You can use a data table to perform a free query, for example, JOIN, GROUP BY, and so on. You can also manually set the data table columns.

Procedure

1. To add a data table, in the *Toolbox* window, *Data Source* area, double-click the *Data Tables* icon.

A new data table is created with a default unique ID and a type.

Note

You can change the default ID and type of the data table. However, changing the type results in the loss of the existing query or columns.

2. To set queries, change the data table type to Query, and in the [Query](#) property field, specify your query. If the data table is attached to a grid control, the grid is automatically filled with columns from the query.
3. To set columns for manual data tables, change the data table type to Manual and in the [Columns](#) property field, click the  button.

In the [CollectionEditorForm](#) window, you can modify the columns:

- o To add a new column, choose the [Add](#) button.
A new column is created with a default name and type (ft_Alphanumeric). You can change the column default name and type.
 - o To delete the column, choose the [Remove](#) button.
4. To bind a data table (with Manual type) to an edit text control, perform the following steps:
 1. In the [Properties](#) window of the edit text control, set the [DataSource](#) property to the data table.
 2. In the [Alias](#) property, set the column of the data table.

3.3.4 Working with Matrixes

The matrix control displays data in a table. You specify each column manually and bind each column to a data source.

Procedure

1. To add a matrix control, in the [Toolbox](#) window, [UI Controls](#) area, double-click the [Matrix](#) icon.
The matrix control is added to the form with a default column Col_0.
2. In the [Properties](#) window, under the [Specific](#) area, in the [Columns](#) property field, click the  button.
The [CollectionEditorForm](#) window appears.
3. Choose the [Add](#) button to add new columns.
4. In the [Properties](#) area, specify the [Type](#) field of the column, the [DataSource](#), [Alias](#) fields and the specific fields.
5. Add as many columns as needed, and choose the [OK](#) button.
6. To preview the matrix, click  ([Preview@B1Client](#)).

3.3.5 Working with Choose From Lists

ChooseFromList is a basic functionality in the SAP Business One application. It allows you to open a ChooseFromList form, which displays a list of objects of the same type.

Procedure

1. To add a ChooseFromList control, in the *Toolbox* window, *Data Source* area, double-click the *Choose From List* icon.

A new ChooseFromList is created with a default unique ID. You can change the default ID.

2. In the *Properties* window, specify the *ObjectType* property, and enter the type of the object you want the Choose from List form to display. For example, enter 2 for business partners.

You can connect *EditText*, *Button*, and *Matrix/Grid* column (with type `it_EDIT` or `it_LINKED_BUTTON`) items to a choose from list.

3. To connect *EditText*, *Button*, and *Matrix/Grid* column items to a choose from list, in the *Properties* window, specify the *ChooseFromListUID* field.
4. In the *ChooseFromListAlias* (*EditText*, *Matrix/Grid* column with type `it_EDIT` or `it_LINK_BUTTON`) field, enter the alias of the field you want to display.

3.3.6 Example: Creating a Business Partner Field with Linked Button and Choose From List Functions

The linked button function is often used together with the choose from list function. For example, you add a new field named *BP Code*. For this field, you want to have a choose from list to open the *List of Business Partners* window and a linked button to open the *Business Partner Master Data* window.

Procedure

1. In the *Toolbox* window, *UI Controls* area, add a *StaticText*, a *LinkedButton* and an *EditText*.



2. In the *Data Source* area, add a *DB Data Source* and a *Choose From List*.
3. In the *Properties* window of the form, specify a valid value for the *ObjectType* field.
4. Set the *Choose From List* properties.

In the *Properties* window of the choose from list, in the *ObjectType* field, enter 2, which represents business partners. For more information about other object types, see *BoObjectTypes* Enumeration in SDK Help Center.

5. Set the *DB Data Source* properties.

In the *Properties* window of the DB data source, in the *TableName* field, enter OCRD, which represents the SAP Business One system table for business partners. For more information about other system tables, see *Database Tables Reference* in SDK Help Center.

6. Connect the linked button with the *BP Code* edit text field.

In the *Properties* window of the linked button, in the *LinkTo* field, paste the unique ID of the edit text; in the *LinkedObject* field, enter 2.

7. Bind the DB data source to the *BP Code* edit text field.

In the *Properties* window of the edit text, in the *DataSource* field, enter OCRD; in the *Alias* field, enter CardCode.

8. Connect the choose from list with the *BP Code* edit text field.

-
- In the *Properties* window of the edit text, in the *ChooseFromListUID* field, enter the ID of the choose from list; in the *ChooseFromListAutoFill* field, select True; and in the *ChooseFromListAlias* field, enter CardCode.
9. To preview, click  (*Preview@B1Client*).
- o When you click the choose from list icon of the *BP Code* field, the *List of Business Partners* window appears.
 - o After you select a business partner from the list, the *BP Code* field is automatically filled and a linked button appears.
 - o Click the linked button; the *Business Partner Master Data* window appears.

3.4 Adding, Renaming or Deleting Forms in a Project

To add forms in your project, right-click your project in the *Solution Explorer* window.

From the context menu, you can add the following types of forms:

- [New user form](#)
- [Existing user form](#)
- [System form](#)
- [UDO form](#)

To rename a form in your project, in the *Solution Explorer* window, right-click the form and choose [Rename](#).

To delete a form in your project, in the *Solution Explorer* window, right-click the form and choose [Delete](#).

3.4.1 Adding New User Forms

A user form is a form designed by a partner. It can be generated from SAP Business One Studio, Screen Painter, UI API or your own code.

To add a new user form, in the *Solution Explorer* window, right-click your project, and choose [New Form](#).

A new form opens in edit mode and appears in the *Solution Explorer* window.

You can now design the new form. For more information, see [Designing a Form](#).

3.4.2 Adding Existing User Forms

You can add an existing user form to your project.

Procedure

1. In the *Solution Explorer* window, right-click your project, and choose Add → Existing Form.
2. In the Add Existing Item window, select the form you want to add.
3. Choose Open.

The form opens in edit mode and appears in the *Solution Explorer* window.

3.4.3 Adding System Forms

A system form is a standard SAP Business One form that you can add to your project.

Procedure

1. In the *Solution Explorer* window, right-click your project, and choose *Add* → *System Form*.
2. In the *Add System Form* window, select the form you want to add.

 Note

The *Add System Form* window lists the following system forms:

- o All currently opened system forms, if SAP Business One is running
- o The locally cached system forms

3. Choose *OK*.

The form opens in edit mode and appears in the *Solution Explorer* window.

 Note

Once opened in edit mode, a system form will be cached up locally.

4. You can now modify the form.

In edit mode of the system form, you can add new UI controls and modify the properties of the existing system items.

5. To preview the system form, you need to open the form in the SAP Business One client first, and then click  (*Preview@B1Client*).

 Caution

Deleting existing system items does not take effect when you preview the modified system form.

3.4.4 Adding UDO Forms

A UDO form is a form generated by the *User-Defined Object Registration* wizard. (To access the wizard, from the SAP Business One menu bar, choose *Tools* → *Customization Tools* → *Objects Registration Wizard*.) You can add a UDO form to your project.

Procedure

1. In the *Solution Explorer* window, right-click your project, and choose *Add* → *UDO Form*.
2. Log in.

For SAP Business One 10.0 FP 2305 or later versions, if you have registered and enabled identity providers in the SLD control center, you can log in with the accounts of the authentication server users, domain users or

the external IDP users that you have bound to the SAP Business One company users. The login scenario depends on the identity providers you activated. For more information, please refer to the guide *Identity and Authentication Management in SAP Business One* on the [SAP Help Portal](#).

For a version earlier than 10.0 FP 2305, in the *Company Logon* window, specify the details of the company you want to work with, and then choose *Log On*.

Field	Description / Activity
<i>Server</i>	Specify the IP address or the machine name of the SAP Business One server.
<i>Server Type</i>	From the dropdown list, select a server type: MSSQL_2008, MSSQL_2012, or HANADB.
<i>License Server</i>	Specify the license server.
<i>Company Name</i>	Specify the company defined on the selected SAP Business One server.
<i>User ID</i>	Specify your user name for the company you want to log on to.
<i>Password</i>	Enter your password.

The UDO form that is registered appears in the *List of UDO Forms* window.

1 Note

Only the Header Line Style (new UDO style in SAP Business One 8.82) is supported. The Matrix Style (old UDO style) is not supported.

3. In the *List of UDO Forms* window, select the UDO form you want to add, and then choose the *Open* button.

The form opens in edit mode and appears in the *Solution Explorer* window.

1 Note

You can add more UDO forms. The application remembers the company logon information, so you do not need to specify it again.

If you restart SAP Business One Studio, you only need to fill out the *Password* field.

4. You can now modify the UDO form.
5. To preview the form, click  (*Preview@B1Client*).
6. To save the changes, in the *Solution Explorer* window, right-click the form and choose *Save to DB*.

When you reopen the form in the SAP Business One client, you can see the UDO form is modified.

1 Note

When you add the UDO forms and fail to complete the action, you can refer to the SAP Note 3332104.

3.5 Editing Active Forms in SAP Business One

In the SAP Business One client, you can edit the active system form through SAP Business One Studio or SAP Business One Studio for Microsoft Visual Studio. The active form means the current system form in the SAP Business One client. If you have multiple forms open in the SAP Business One client, the active form is the top one.

Procedure

1. In SAP Business One, from the [Tools](#) menu, choose [SAP Business One Studio](#).
2. If you want to edit the active form from SAP Business One Studio, choose [Edit Active Form](#).
 - o If SAP Business One Studio is already running, the application adds the active system form to the project.
 - o If you have not yet opened SAP Business One Studio, the application runs SAP Business One Studio first, automatically creates a project, and opens the active form in the project.
3. If you want to edit the active form from SAP Business One Studio for Microsoft Visual Studio, choose [Edit Active Form in Visual Studio](#).
 - o If the SAP Business One Studio for Microsoft Visual Studio is already running, the application adds the active system form to the project.
 - o If you have not yet opened Microsoft Visual Studio, the application does nothing for you. You need to open and create a project in Microsoft Visual Studio first, and then in the SAP Business One client, from the [Tools](#) menu, choose [SAP Business One Studio](#), and choose [Edit Active Form in Visual Studio](#).

3.6 Exporting Updated Parts of a System Form

If you edited a system form, you can export the updated part of the system form to a .sr f file.

Procedure

1. In the Solution Explorer window, right-click the system form and choose Export Updated Part As.
2. In the Export File As window, specify the file name.
3. Choose Save.

4 Working with Add-on Projects in SAP Business One Studio for Microsoft Visual Studio

4.1 Creating SAP Business One Add-on Projects

You can create your new add-on forms with the same look and feel as the SAP Business One forms.

Procedure

1. In the Microsoft Visual Studio main window, choose *New* → *Project*.
After you install SAP Business One Studio for Microsoft Visual Studio, you can locate the *SAP Business One* template in the *New Project* dialog box by navigating the expanding list in the left pane under *Installed Templates*.
2. In the *New Project* window, from the *Installed Templates* area, select the *SAP Business One* template.
3. To create an add-on project with an empty user form, select *SAP Business One Add-on Project* and specify the name and the location of your new project.
4. To create an add-on project using an existing B1s file from SAP Business One Studio, select *SAP Business One Add-on Project from B1s File*.
5. Choose *OK*.

A default form named *Form1.b1f* appears in the *Solution Explorer* window, and the layout of the form appears in the design area. A folder *B1Studio* is generated under
C:\Temp\<YourAddOnProject>\<YourAddOnProject>\bin\Debug.

You can design the layout of your form and the logic of the controls. For more information, see [Designing a Form](#).

Note

When you create an SAP Business One add-on project, it adds the *SAPbouiCom.Framework*, which is for connecting the UI API application, handling *LoadBatchAction*, system/user/UDO form events, and generating code. It includes:

- o *Menu.cs* - for adding menus and handling menu events
- o *Programs.cs* - for connecting the UI server and performing the initialization
- o *SAPbouiCOM.Framework.Application.SBO_Application* - the root COM object of UI API
Sample Code: `Application.SBO_Application.MessageBox("Hello world");`
- o *SAPbouiCOM.Framework.FormBase.UIAPIRawForm* - the original form COM object of UI API
Sample Code: `this.UIAPIRawForm.Close();//Close form`

The *SAPbouiCom.Framework* contains all UI API COM objects and interfaces; therefore, we do not recommend that you replace it.

Note

In SAP Business One 9.1, a new .net SAPBusinessOneSDK.dll is added. You can build add-on solutions using SAPBusinessOneSDK.dll, which integrates UI API and DI API.

4.2 Tool Windows in SAP Business One Studio for Microsoft Visual Studio

The tool windows in SAP Business One Studio for Microsoft Visual Studio are very similar to the tool windows in SAP Business One Studio. The windows include:

- **Toolbox** window – contains the controls you can use to design your form. For more information, see [Toolbox Window](#).

Note

To program a certain control, double-click it. The code is automatically generated.

- **UI Outline** window – allows you to view items in a selected pane.

Note

This window may not appear by default. To display the **UI Outline** window, from the menu bar, choose **View** → **Other Windows** → **Outline Window**.

- **Solution Explorer** window – allows you to view the source files in your project.

Note

The project files also include SAP Business One UI API references and predefined menu options (Menu.cs). In runtime, by default, the newly added form appears at the bottom of the SAP Business One **Main Menu**. You can change the code if you want to open the form from another menu location.

- **Properties** window – allows you to set and examine the properties and events of forms and form items.

Note

To register events of a control, in the **Properties** window, click  (**Events**), and double-click the event property you want to edit. The code is automatically generated, and you can program as needed.

You can drag the border of the windows to resize the window, click  (**Auto Hide**) to hide the window, and also dock the windows to a new location (see [Docking Tool Windows](#)).

- The double-click function - you can double-click an item to add an event. The following are the default events that the system supports currently.

Item Type	Event
Form	LoadAfter
EditText	KeyDownAfter
CheckBox	ClickBefore
OptionButton	ClickBefore

Item Type	Event
Button	ClickBefore
ComboBox	ComboSelectAfter

- **Toolbar** - Located below the [Menu](#) button, the toolbar is a collection of icons providing easy access to commonly used functions. The functions are the same as those for SAP Business One Studio. See chapter 2.2 in this document.

4.3 Adding Forms in a Project

To add forms in your project, in the [Solution Explorer](#) window, right-click your project.

From the context menu, choose [Add](#) → [New Item....](#) You can add the following types of forms:

- [SAP Business One user form](#)
- [SAP Business One system form](#)
- [SAP Business One UDO form](#)

4.3.1 Adding SAP Business One User Forms

An SAP Business One user form is a form designed by a partner. It can be generated from SAP Business One Studio, Screen Painter, UI API, or your own code.

Procedure

1. In the [Solution Explorer](#) window, right-click your project and choose [Add](#) → [New Item....](#)
2. In the [Add New Item](#) window, from the [Installed Templates](#) area, select the [SAP Business One](#) add-on template.
3. Choose [SAP Business One User Form](#).

A new form opens in edit mode and appears in the [Solution Explorer](#) window.

You can now design the new form. For more information, see [Designing a Form](#).

In runtime, by default, the newly added form appears at the bottom of the SAP Business One [Main Menu](#).

4.3.2 Adding SAP Business One System Forms

A system form is a standard SAP Business One form that you can add to your project.

Procedure

1. In the [Solution Explorer](#) window, right-click your project, and choose [Add](#) → [New Item....](#)
2. In the [Add New Item](#) window, from the [Installed Templates](#) area, select the [SAP Business One](#) add-on template.
3. Choose [SAP Business One System Form](#).

Note

The [Add System Form](#) window lists the following system forms:

- o All currently opened system forms, if SAP Business One is running
- o The locally cached system forms

4. Choose [OK](#).

The form opens in edit mode and appears in the [Solution Explorer](#) window.

You can add user items, modify the properties of a system item, and register events, but you cannot delete existing system items. The modification is implemented after you reopen the system form.

4.3.3 Adding SAP Business One UDO Forms

A UDO form is a form generated by the [User-Defined Object Registration](#) wizard. (To access the wizard, from the SAP Business One menu bar, choose [Tools](#) → [Customization Tools](#) → [Objects Registration Wizard](#).) You can add a UDO form to your project.

Procedure

1. In the [Solution Explorer](#) window, right-click your project, and choose [Add](#) → [New Item....](#)
2. In the [Add New Item](#) window, from the [Installed Templates](#) area, select the [SAP Business One](#) add-on template.
3. Choose [SAP Business One UDO Form](#).
4. Log in.

For SAP Business One 10.0 FP 2305 or later versions, if you have registered and enabled identity providers in the SLD control center, you can log in with the accounts of the authentication server users, domain users or the external IDP users that you have bound to the SAP Business One company users. The login scenario depends on the identity providers you activated. For more information, please refer to the guide [Identity and Authentication Management in SAP Business One](#) on the [SAP Help Portal](#).

For a version earlier than 10.0 FP 2305, in the [Company Logon](#) window, specify the details of the company with which you want to work, and choose [Log On](#).

Field	Description / Activity
Server	Specify the IP address or the machine name of the SAP Business One server.
Server Type	From the dropdown list, select a server type: MSSQL_2008, MSSQL_2012, or HANADB.

Field	Description / Activity
License Server	Specify the license server.
Company Name	Specify the company defined on the selected SAP Business One server.
User ID	Specify your user name for the company to which you want to log on.
Password	Enter your password.

The UDO form that is registered appears in the [List of UDO Forms](#) window.

Note

Only the Header Line Style (new UDO style in SAP Business One 8.82) is supported. The Matrix Style (old UDO style) is not supported.

- In the [List of UDO Forms](#) window, select the UDO form you want to add, and then choose the [Open](#) button.
The form opens in edit mode and appears in the [Solution Explorer](#) window.

Note

The application remembers the company logon information, so if you want to add more UDO forms, you do not need to specify that information again.

If you restart SAP Business One Studio, you only need to fill out the [Password](#) field.

- You can now modify the UDO form.
To save the changes, in the [Solution Explorer](#) window, right-click the form and choose [Save UDO to DB](#).
When you reopen the form in the SAP Business One client, you can see the UDO form is modified.

4.4 Loading an Existing B1s File to Your Project

You can load your existing .b1s solution or .srf file (ScreenPainter file) in the following ways:

Creating an Add-on Project Using an Existing B1s File

- In the Microsoft Visual Studio main window, choose [New](#) → [Project](#).
- In the [New Project](#) window, from the [Installed Templates](#) area, select the [SAP Business One](#) add-on template.
- To create an add-on project using an existing B1s file from SAP Business One Studio, select [SAP Business One Add-on Project from B1s File](#).
- Choose [OK](#).

Adding an Existing B1s File to the Current Project

- In the [Solution Explorer](#) window, right-click your project and choose [Add](#) → [Existing Business One File](#).
- In the [Open](#) window, select the [Business One Studio files \(*.b1s\)](#) file you want to add.
- Choose [Open](#). The form opens in edit mode and appears in the [Solution Explorer](#) window.

Adding an Existing SRF File to the Current Project

1. In the *Solution Explorer* window, right-click your project and choose *Add* → *Existing Business One File*.
2. In the *Open* window, select the *Business One Studio SRF files (*.srf)* file you want to add.
3. Choose *Open*. The form opens in edit mode and appears in the *Solution Explorer* window.

4.5 Packaging Your Project

The SAP Business One Extension Package features are integrated in SAP Business One Studio for Microsoft Visual Studio. You can package your project directly from the menu bar.

Procedure

1. Build your project first to get the executable file of your extension.
2. From the menu bar of the Microsoft Visual Studio main window, choose *SAP Business One Studio* → *Extension Package*.
3. In the *Extension Registration Data Generator* window, expand *Basic Information*, and specify the following fields:
 - o *Extension Name* – Enter the name of the extension. This field is mandatory.
 - o *Extension Version* – Enter the version of the extension for which you want to package and generate the ARD file. This field is mandatory.
 - o *Extension Provider* – Enter the name of the SAP partner that creates and owns the extension. For example, the name of your company.
 - o *Extension Namespace* – Enter a name for the folder in which SAP Business One places the extension after a user registers the extension in the application.
 - o *Supported Database* – Specify the database in which the extension works. This field is mandatory.
 - o *Contact Data* – Enter contact information for the SAP partner that creates and owns the extension. For example, enter the URL of your company's Website.
 - o *Supported Client Type* - Specify this field according to your own add-on strategy. You may design your add-on to run in Windows desktop mode and in Browser Access mode (using the *Both* type). Also, you may create two add-on versions, one for Windows desktop mode (using the *Desktop* type), and another for Browser Access mode (using the *Browser* type).

Note

If you have not yet repackaged your add-ons using this Extension Package tool with the *Supported Client Type* field, your existing add-ons are compatible with the *Desktop* type.

- o *Multiversion* - For SAP Business One Cloud only, if you select this field, the application treats add-on 1.0 and add-on 2.0 as two different add-ons, while not treats add-on 2.0 as an upgrade version of add-on 1.0.
4. Expand *Extension File*, specify the path of the executable file of your 32-bit or 64-bit add-on, and select the files that should be packaged in the zip file.

To package your apps for the version for SAP HANA, specify the app zip file, the name and the package of your app.

Note

The naming convention for your package is *.*. For example, if the package hierarchy is sap.sbo.atp in SAP HANA, the package name is this, and the corresponding URL will be <https://host:port/sap/sbo/atp>.

5. Expand [Deployment Steps](#), and select the COM DLLs to register for your 32-bit or 64-bit add-on.
6. Expand [SBO Compatibility](#); you can specify the versions of SAP Business One with which the add-on is compatible.

In the [Compatible with SAP Business One](#) area, specify the following:

- o [Version From](#) – Enter the earliest version of SAP Business One with which the extension is compatible.
- o [To](#) – Enter the latest version of SAP Business One with which the lightweight add-on is compatible.

Example

Enter compatible versions of SAP Business One with the format **<xxx.yyy.xx>**, for example, **910.000.00**.

7. Expand [Parameters](#); you can optionally specify shared parameters and parameters for the extension.
 - o Shared Parameters - The parameters or configuration required to run the extension. The parameters are shared in all extension instances running on different companies.
 - o Parameters - The parameters or configurations required to run the extension.
8. Choose the [Package](#) button.
9. In the [Save As](#) window, specify the location where you want to save the file and choose the [Save](#) button.

Note

If you are not yet ready to package your extension, you can choose the [Save](#) button instead of the [Package](#) button. The information you entered in the [Extension Registration Data Generator](#) window will be saved in your project. The next time you open the [Extension Registration Data Generator](#) window, the saved information will be displayed.

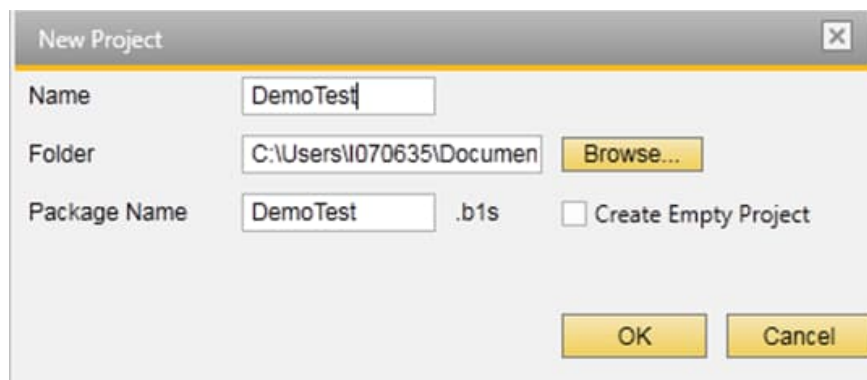
For more information, see *How to Package and Deploy SAP Business One Extensions for Lightweight Deployment*.

5 SAP Business One Studio Suite Demo Script

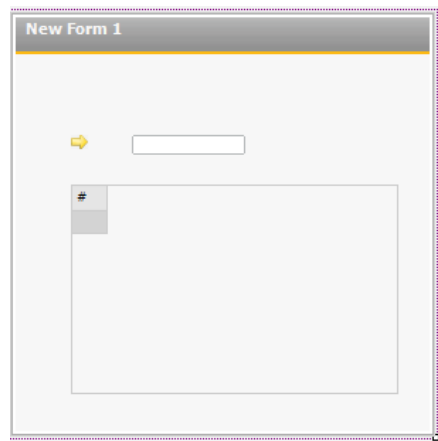
This section introduces a demo for how to work with your add-on projects in SAP Business One Studio Suite.

Procedure

1. Start SAP Business One and log on to a company.
 - o Database: MadridDemo
 - o User: manager
 - o Password: 1234
2. Start SAP Business One Studio and add a new project.
 1. Choose [Menu](#) → [New](#) → [Project](#) → [Add-On](#).
 2. Choose [OK](#).

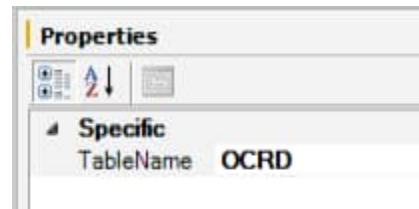


3. Create a user form; add an [EditText](#) with [ChooseFromList](#); and add a grid with [DataTable](#).
 1. Add EditText, Grid, LinkButton, DB DataSource, Data Table, and ChooseFromList.

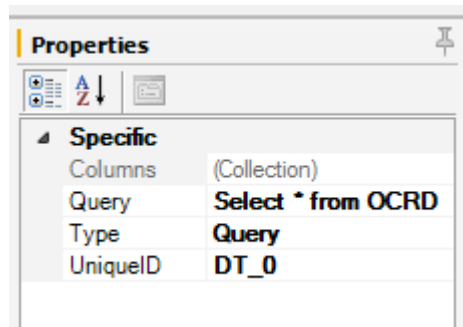


2. Specify the controls properties as follows:

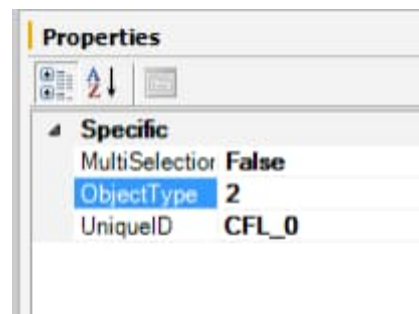
- o DB DataSource:



- o Data Table:



- o ChooseFromList:

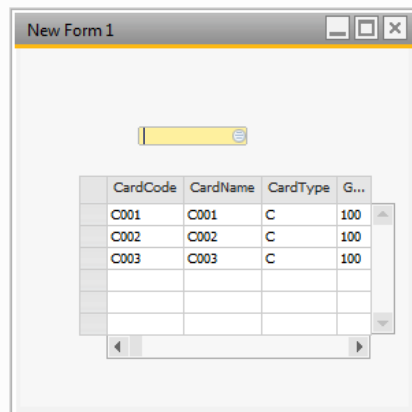


- o EditText:

DataBind		
Alias	CardCode	ChooseFromList CardCode
DataBound	True	ChooseFromList True
DataSource	OCRD	ChooseFromList CFL_0

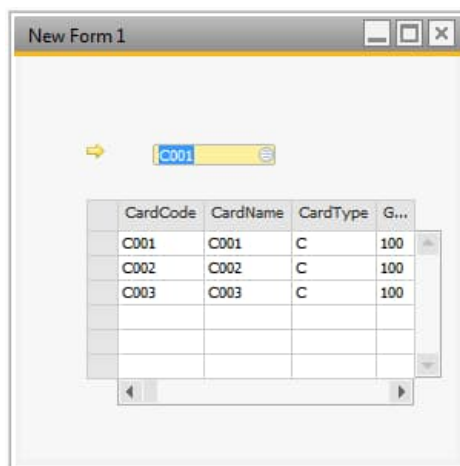
- o LinkButton: *LinkTo* = **Item_0**; *LinkedObject* = **2**
 - o Grid: *DataTableID* = **DT_0**
 - o Form: *LinkedObject* = **2**
3. From *Toolbar*, choose *Preview* .

The user form opens in the SAP Business One client. The grid displays all BPs.



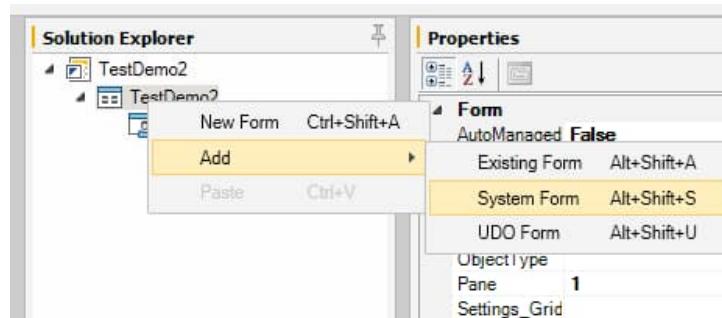
CardCode	CardName	CardType	G...
C001	C001	C	100
C002	C002	C	100
C003	C003	C	100

4. To select a business partner, choose the *ChooseFromList* icon. The application fills the business partner code in the EditText,
To open the business partner form, choose the linked button.

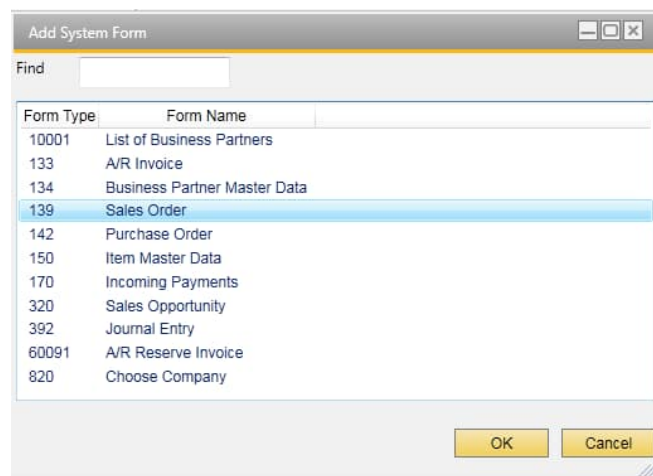


CardCode	CardName	CardType	G...
C001	C001	C	100
C002	C002	C	100
C003	C003	C	100

4. Edit a system form (sales order); add a button and preview it.
1. In the SAP Business One client, from the *Main Menu*, choose *Sales* → *Sales Order* to open the original sales order.
 2. Right-click the project node to add a system form.



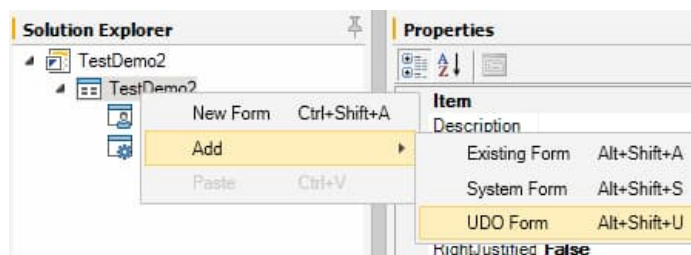
3. Select *Sales Order* and choose *Open* (the other form is cache file in local that user opened).



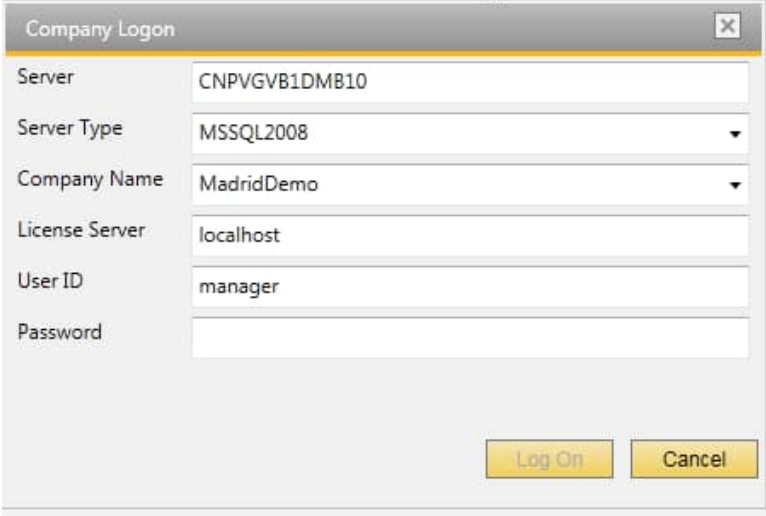
4. From *Toolbar*, add the button and choose *Preview*.
5. Switch to the SAP Business One client to review the updates of the sales order form.

5. Edit a UDO form; add a button; and save the UDO to the database.

1. To open the original UDO form, in the SAP Business One client, choose **Tools** → **Default Forms** → **UDO1-UDO1**.
2. To add a UDO form, right-click the project node.



3. Specify the DI connection information (password: 1234) and choose **Log On**.



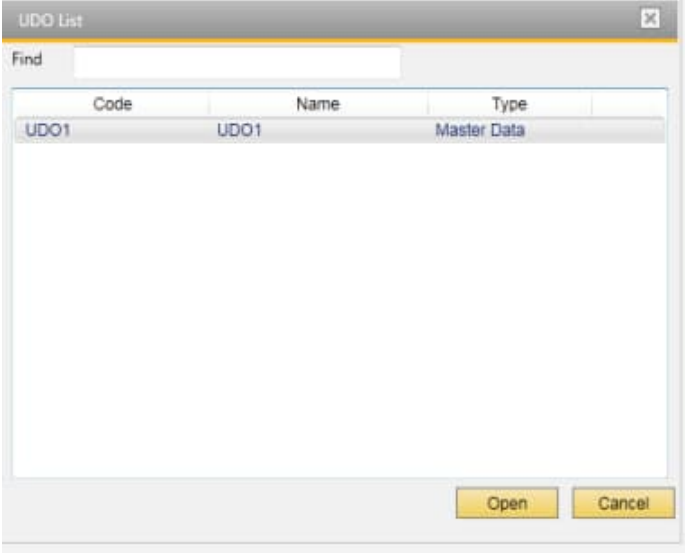
The 'Company Logon' dialog box contains the following fields and values:

Server	CNPVGVB1DMB10
Server Type	MSSQL2008
Company Name	MadridDemo
License Server	localhost
User ID	manager
Password	

Buttons: Log On, Cancel

As of SAP Business One 10.0 FP 2305, if you have registered and enabled identity providers in the SLD control center, you can log in with the accounts of the authentication server users, domain users or the external IDP users that you have bound to the SAP Business One company users. The login scenario depends on the identity providers you activated. For more information, please refer to the guide *Identity and Authentication Management in SAP Business One* on [SAP Help Portal](#).

4. To edit the UDO form, select **UDO1** and choose **OK**.

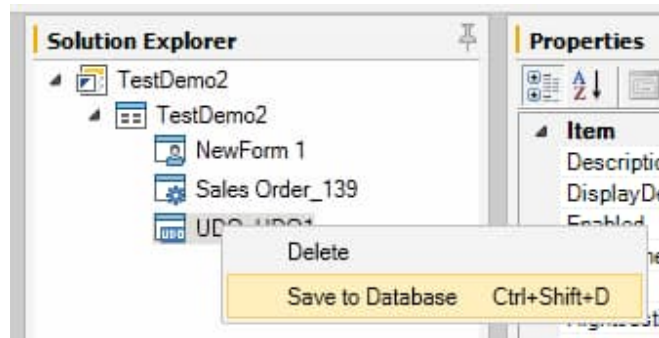


The 'UDO List' dialog box shows a table with the following data:

Code	Name	Type
UDO1	UDO1	Master Data

Buttons: Open, Cancel

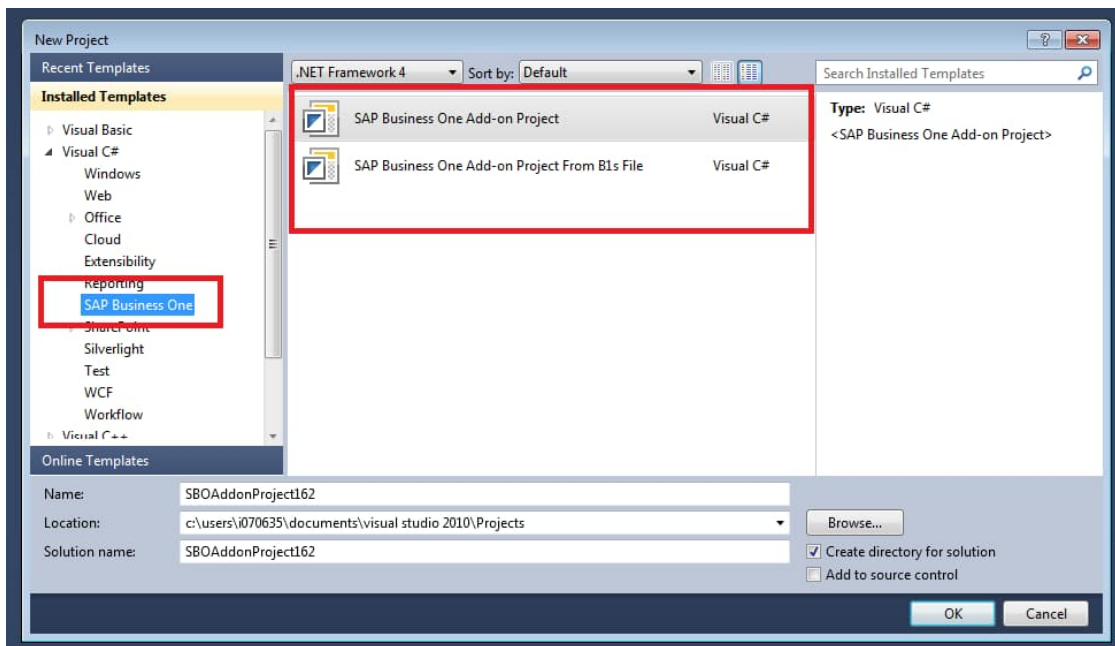
5. Add the button and save the UDO to the database.



6. To review the updates, from the SAP Business One client, reopen the UDO form.

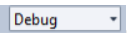
The screenshot shows the 'UD01' form in SAP Business One Studio. The form has two main sections. The top section contains fields for 'Code', 'DocEntry', 'Object', 'UserSign', 'CreateDate', 'UpdateDate', 'Name', 'Canceled', 'LogInst', 'Transferred', 'CreateTime', and 'UpdateTime'. The bottom section contains a table with columns '#', 'Code', 'LineId', 'Object', and 'LogInst'. The table has one row with 'LineId' value '1'. Below the table are buttons for 'Add', 'Cancel', and 'Button2'.

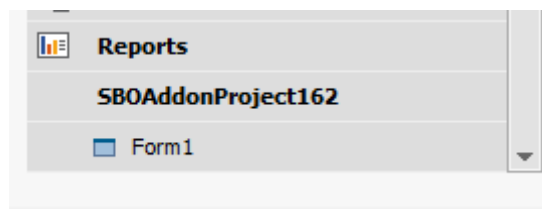
6. Start SAP Business One Studio for Microsoft Visual Studio and add a new project.
 1. Choose [Menu](#) → [File](#) → [New](#) → [Project](#) and select the C# project templates.
 - o The template [SAP Business One Add-on Project](#) is for creating a new project.
 - o The template [SAP Business One Add-on Project from B1s File](#) is for creating a project with the existing B1s file.



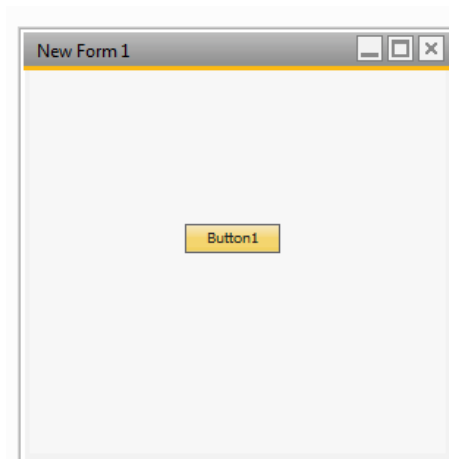
2. Select *SAP Business One Add-on Project* and create a new project.
7. Add a button, a Button_ClickBefore event, and a popup message box if choosing the button.
 1. Add a button.
 2. Double-click the button to add a Button_ClickBefore event.
 3. Add the UI API code as follows.

```
private void Button0_ClickBefore(object sboObject, SAPbouiCOM.SBOItemEventArgs pVal, out bool BubbleEvent)
{
    BubbleEvent = true;
    Application.SBO_Application.MessageBox("Hello World");
}
```

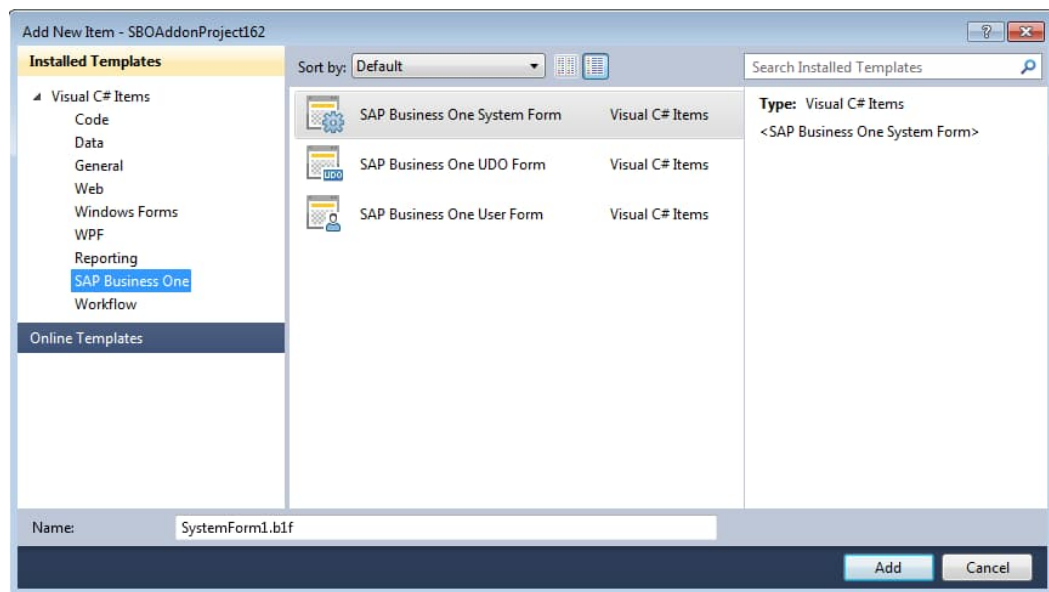
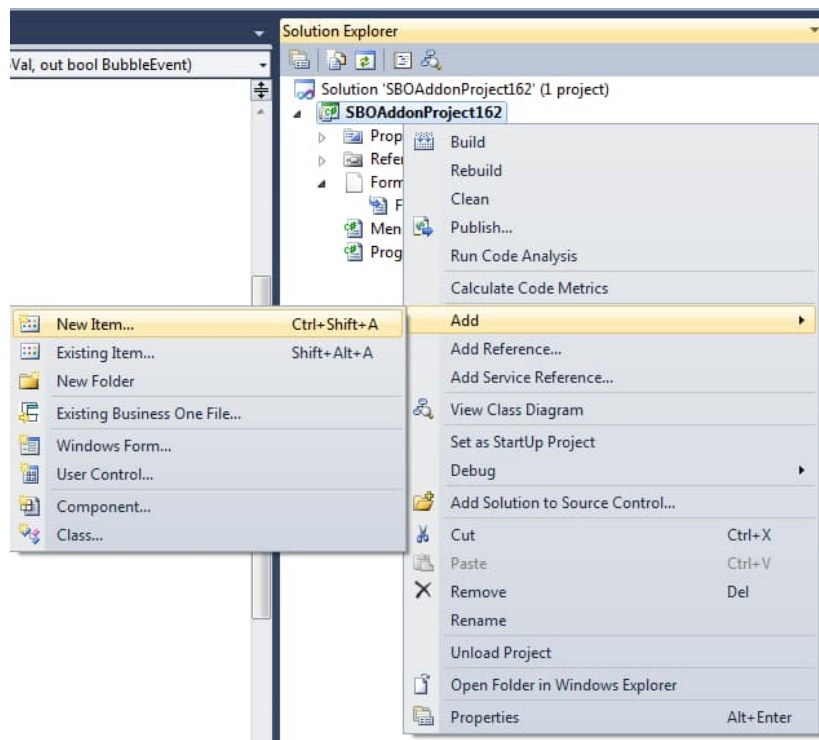
4. To run the add-on, from *toolbar*, or by using the F5 shortcut, choose *Debug*  .
- In the *Main Menu* of the SAP Business One client, the add-on adds the user menu.
5. Choose the menu to open the user form.



6. Choose the button to open a message box.

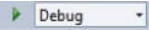


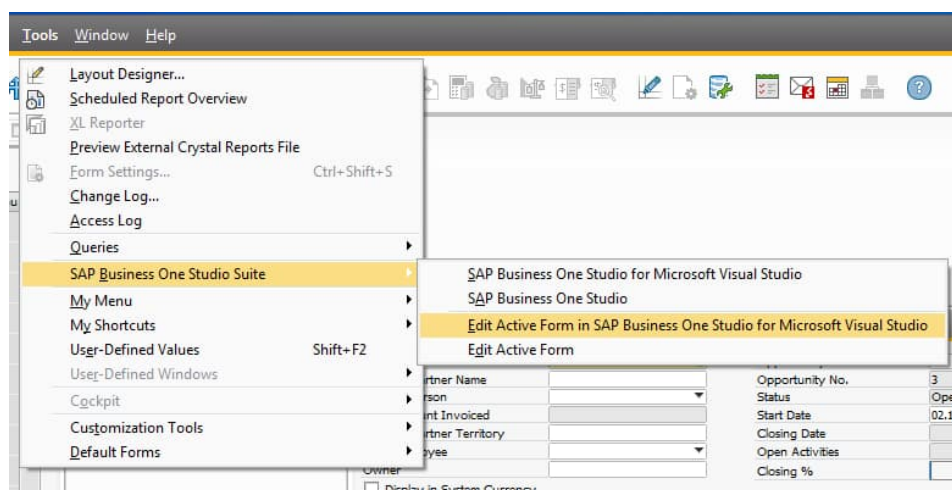
7. Stop debugging and close all forms.
8. Edit a system form (sales order). Add a button and a Button_ClickBefore event; create a user form if choosing the button.
 1. In the SAP Business One client, from the *Main Menu*, choose *Sales* → *Sales Order* to open the original sales order.
 2. Right-click the project node to add a system form (sales order).



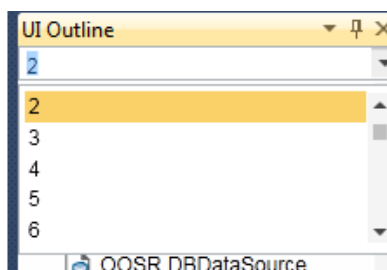
3. Add a button.
4. Double-click the button to add a Button_ClickBefore event.
5. Add the UI API code as follows:

```
private void Button26_ClickBefore(object sboObject, SAPbouiCOM.SBOItemEventArgs pVal, out bool BubbleEvent)
{
    BubbleEvent = true;
    Form1 activeform = new Form1();
    activeform.Show();
}
```

6. To run the add-on, from [toolbar](#), or by using the F5 shortcut, choose [Debug](#) .
7. From the SAP Business One client, open the sales order. The new button appears.
To open the new form, choose the button.
9. Edit a system form (sales opportunity) from the menu; design the UI API code to set [Remark](#) as a mandatory field.
 1. In the SAP Business One client, from the [Main Menu](#), choose [Sales Opportunities](#) → [Sales Opportunity](#) to open the original sales opportunity form.
 2. From the SAP Business One client, choose [Edit Form in Visual Studio](#) to edit the system form.

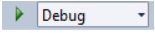


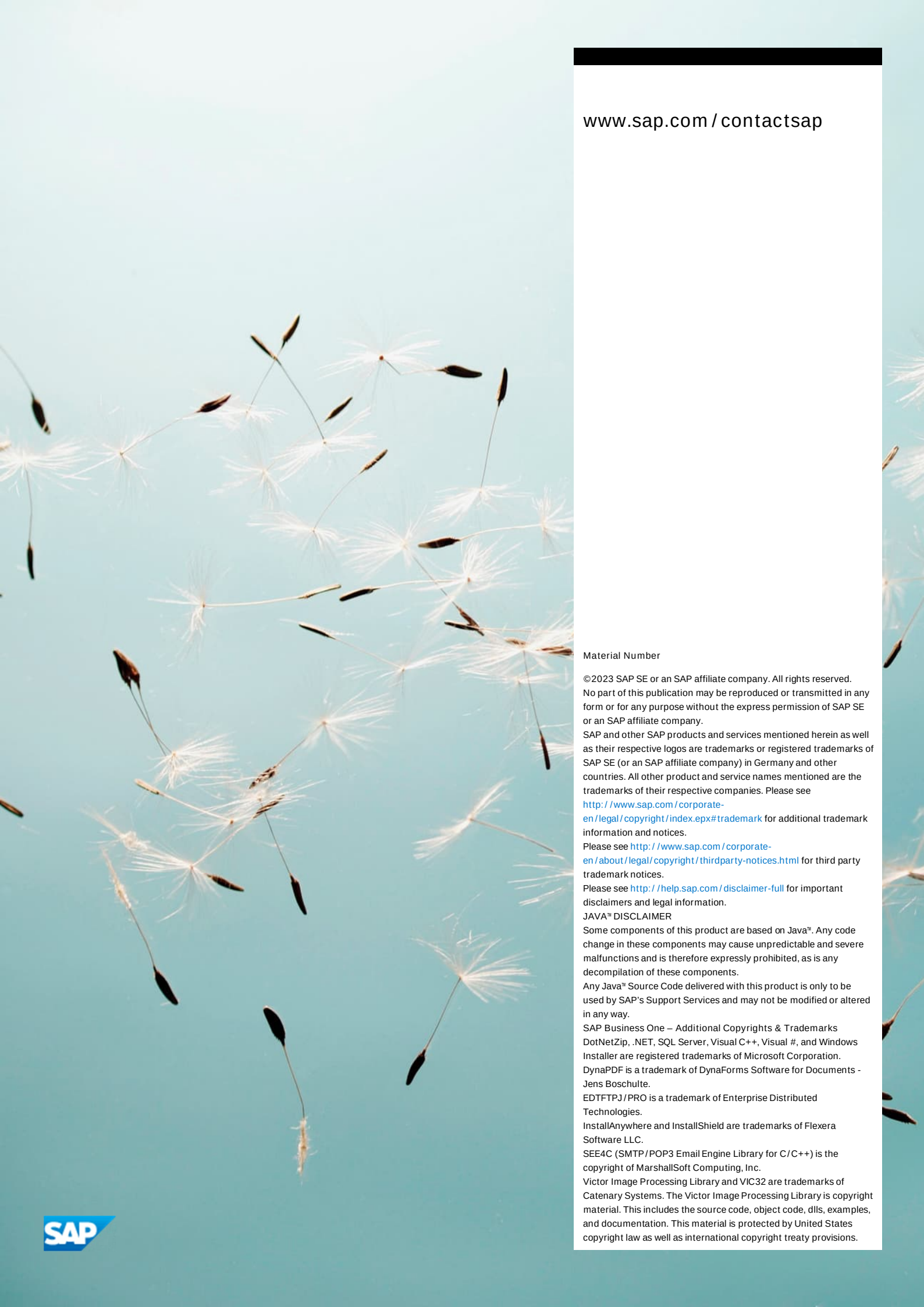
3. In [UI Outline](#), change the pane level to 2.



4. Add a button.
5. Double-click the button to add a Button_ClickBefore event.
6. Add the UI API code as follows:

```
private void Button8_ClickBefore(object sboObject, SAPbouiCOM.SBOItemEventArgs pVal, out bool BubbleEvent)
{
    BubbleEvent = false;
    if (EditText23.String.Length == 0)
    {
        Application.SBO_Application.SetStatusBarMessage("Remark is mandatory.", SAPbouiCOM.BoMessageTime.bmt_Long, true);
    }
    else
    {
        BubbleEvent = true;
    }
}
```


-
7. To run the add-on, from *toolbar*, or by using the F5 shortcut, choose *Debug* .
 8. In the SAP Business One client, reopen the sales opportunity.
 9. On the *Potential* tab, specify the business partner code and potential amount.
On the *General* tab, leave the *Remark* field empty,
 10. Choose *Add*,
In the status bar, the error message: *Remark is mandatory* appears.
 11. Specify the remark. The system adds the sales opportunity.



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